MODULE 4

Electrolyser Technology Deep Dive

Cell → stack → system · AEL · PEM · SOEC · AEM

4.1 Basic concept: the electrolysis cell

Every electrolyser cell has the same five functional layers: a **positive electrode (anode)** where oxygen is formed, a **negative electrode (cathode)** where hydrogen is formed, an **ion conductor** (liquid electrolyte, polymer membrane or ceramic) between them, **porous transport layers** that let water in and gas out, and **bipolar plates** that carry current to the next cell and separate the gases. Cells are stacked in series (like batteries in a torch) to build voltage; stacks are combined in parallel to build capacity.

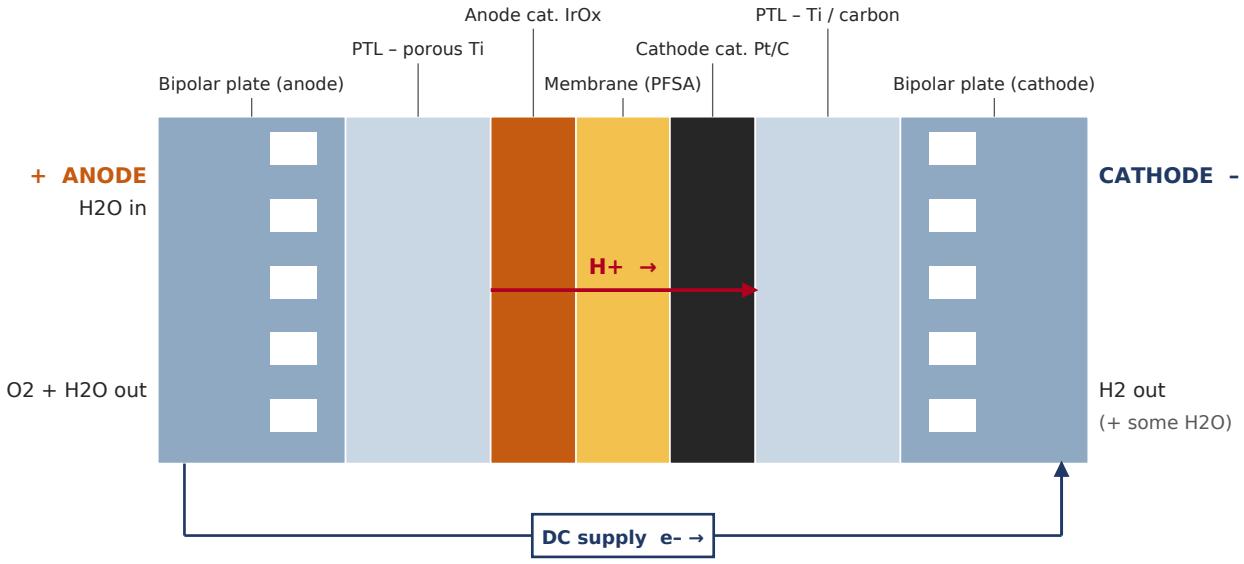


Figure 4.1 – PEM cell cross-section. Water is fed to the anode; protons (H^+) cross the membrane; electrons travel through the external DC circuit; H_2 leaves the cathode. Materials per IRENA Table 1 [R1].

4.2 Electrochemical reactions at each electrode

Type	Anode (+) – oxygen side	Cathode (-) – hydrogen side	Ion carried
Alkaline (AEL)	$4OH^- \rightarrow O_2 + 2H_2O + 4e^-$	$4H_2O + 4e^- \rightarrow 2H_2 + 4OH^-$	OH^- (through porous diaphragm)
PEM	$2H_2O \rightarrow O_2 + 4H^+ + 4e^-$	$4H^+ + 4e^- \rightarrow 2H_2$	H^+ (through PFSA membrane)
AEM	$4OH^- \rightarrow O_2 + 2H_2O + 4e^-$	$4H_2O + 4e^- \rightarrow 2H_2 + 4OH^-$	OH^- (through polymer membrane)
SOEC	$2O^{2-} \rightarrow O_2 + 4e^-$	$2H_2O + 4e^- \rightarrow 2H_2 + 2O^{2-}$	O^{2-} (through YSZ ceramic)

Engineering principle – voltage, current and kilograms

- **Reversible voltage** 1.23 V (25 °C) = minimum electrical work (ΔG). **Thermoneutral voltage** 1.48 V = total energy (ΔH, HHV). Real cells run above 1.48 V; the excess becomes heat that the cooling system removes.
- **Faraday's law:** hydrogen output ∝ current. 2 electrons per H_2 molecule → 1 kg H_2 needs ≈ 26.6 kAh of charge (2 × 96,485 C/mol ÷ 2.016 g/mol).
- **Energy per kg** ≈ 26.6 kAh × cell voltage. At 1.8 V/cell: ≈ 47.9 kWh/kg at stack level (before rectifier and balance-of-plant). This is why lowering cell voltage is the main efficiency lever.
- **Current density** (A/cm²) sets how much H_2 each m² of cell makes – higher density = smaller, cheaper stack but higher voltage.

4.3 Stack construction – exploded view

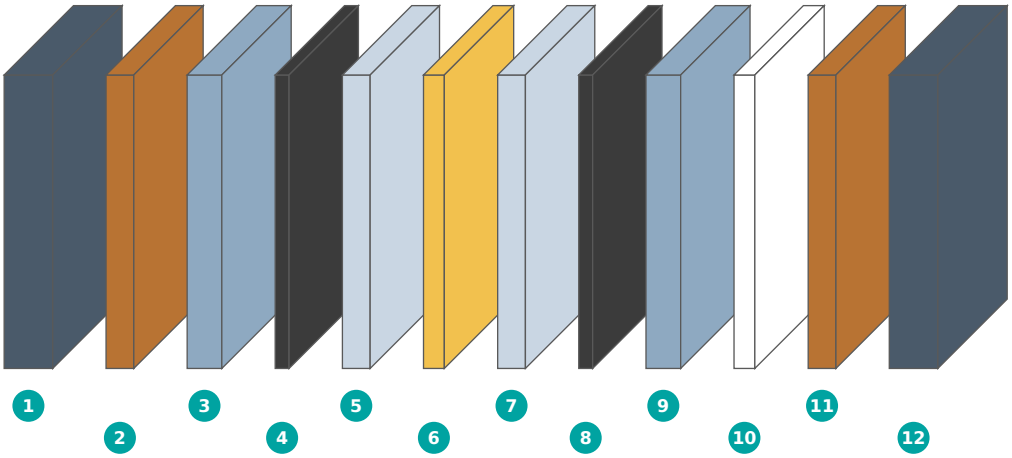


Figure 4.2 – Exploded view of a filter-press stack (numbers match the table).

#	Component	Function	Typical material (IRENA Table 1 [R1])
1	End plate	Clamps the stack; distributes compression	Steel
2	Current collector	Brings DC current into the stack	Copper / nickel-plated
3	Bipolar plate	Conducts current between cells; separates gases; flow channels	AEL/AEM: Ni-coated stainless; PEM: Pt/Au-coated Ti; SOEC: Co-coated stainless (cathode)
4	Seal / gasket	Prevents internal & external leakage	AEL: PSU, PTFE, EPDM; PEM: PTFE, PSU, ETFE; AEM: PTFE, silicone; SOEC: ceramic glass
5	Porous transport layer (anode)	Feeds water, removes O_2, conducts current	AEL: Ni mesh; PEM: Pt-coated sintered Ti; AEM: Ni foam; SOEC: Ni mesh/foam
6	Electrode + membrane (CCM)	Reaction site + ion conductor	PEM: IrO_x anode, Pt/C cathode, PFSA; AEL: Ni-coated steel + ZrO_2/PPS diaphragm; SOEC: LSCF/LSM, YSZ, Ni/YSZ
7	PTL (cathode)	Removes H_2, conducts current	PEM: sintered Ti or carbon cloth; AEM: Ni foam / carbon cloth
8–9	Seal + next bipolar plate	Starts the next cell in series	As above
11–12	Collector + end plate + tie rods	Close the stack; maintain clamping force	Steel tie rods, disc springs

4.4 Balance of plant (system level)

Sub-system	What it does	Key points
Power supply & rectifier	AC→DC at stack voltage/current	Efficiency and harmonics affect total kWh/kg

Sub-system	What it does	Key points
Gas–liquid separation	Separators on H_2 and O_2 sides	Level/pressure balance; alkaline lye return
Water circulation	Feeds water/lye, removes heat & gas	Pumps, filters, ion-exchanger (PEM keeps water ultra-pure)
Cooling system	Removes heat from overvoltage losses	Plate heat exchangers + cooling tower/dry coolers
Gas purification	De-oxo + dryer	Delivers spec purity and dew point
Control system	PLC/DCS, cell-voltage monitoring, SIS	Trips on H_2-in-O_2, pressure, temperature, low flow

4.5 Technology-by-technology

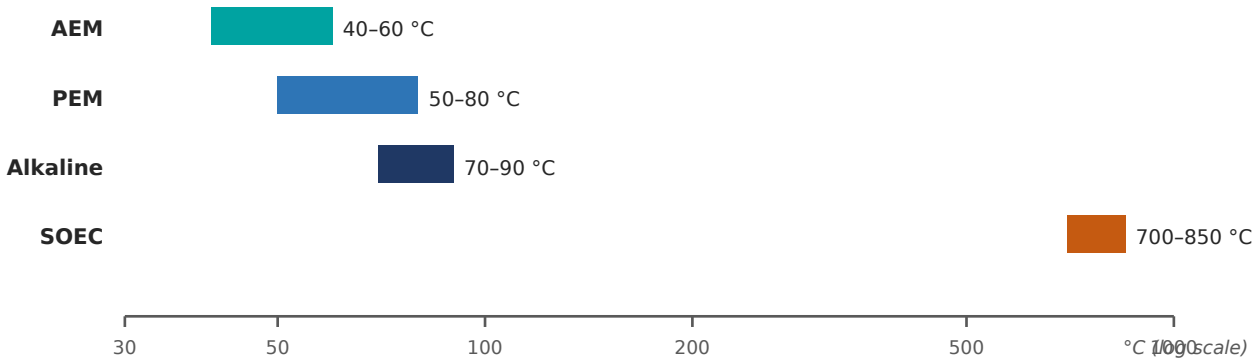
Alkaline (AEL). Simple stack, easy to manufacture; electrode areas up to ~3 m²; concentrated KOH electrolyte; robust ZrO_2-based diaphragms; nickel-coated stainless electrodes. Gas intermixing through the porous diaphragm limits low-load operation and high pressure; modern zero-gap designs have narrowed the gap to PEM. Classic designs have reached lifetimes above 30 years [R1].

PEM. Thin (~0.2 mm) PFSA membrane with advanced electrodes gives higher efficiency and current density, compact footprint and fast dynamic response; it can produce hydrogen at higher pressure. Needs precious metals (iridium, platinum) and titanium components [R1].

AEM. Alkaline chemistry with a solid polymer membrane – aims to combine AEL's cheap materials with PEM's compact design; membrane mechanical stability and durability are still being developed (IRENA classed it lab-scale in 2020) [R1].

SOEC. Ceramic cells at 700–850 °C split steam. Part of the energy can be supplied as heat (e.g. industrial waste heat), giving the highest electrical efficiency; durability and scale are the challenges (lab-scale in IRENA 2020) [R1].

Operating temperature window by electrolyser type (IRENA, 2020)



4.6 Comprehensive comparison (IRENA, 2020)

Parameter	Alkaline	PEM	AEM	SOEC
Operating temperature	70–90 °C	50–80 °C	40–60 °C	700–850 °C
Operating pressure	1–30 bar	<70 bar	<35 bar	1 bar
Electrolyte	KOH 5–7 mol/L	PFSA membrane	DVB polymer + KOH or $NaHCO_3$	Yttria-stabilised zirconia
Separator	ZrO_2 + PPS mesh	Membrane itself	Membrane itself	Ceramic itself
O_2-side catalyst	Ni-coated perforated SS	Iridium oxide	Ni or NiFeCo alloys	Perovskite (LSCF, LSM)
H_2-side catalyst	Ni-coated perforated SS	Pt nanoparticles on carbon	High-surface Ni	Ni/YSZ
System energy, 2020	50–78 kWh/kg	50–83 kWh/kg	57–69 kWh/kg	45–55 kWh/kg

Parameter	Alkaline	PEM	AEM	SOEC
Lifetime, 2020	60,000 h	50,000–80,000 h	>5,000 h	<20,000 h
Stack capex >1 MW, 2020	USD 270/kW	USD 400/kW	–	>USD 2,000/kW
System capex >10 MW, 2020	USD 500–1,000/kW	USD 700–1,400/kW	–	–
2050 targets	<45 kWh/kg; 100,000 h	<45 kWh/kg; 100–120 kh	<45 kWh/kg; 100 kh	<40 kWh/kg; 80 kh
Status (IRENA 2020)	Commercial	Commercial	Lab-scale	Lab-scale

Source: IRENA (2020), *Green Hydrogen Cost Reduction*, Table 1 and Table ES1 [R1]. Both tables report 2020 status – check current OEM data sheets before design work.

Quiz & revision – Module 4

Answer first, then check the key at the end of the module. Score yourself in the Progress Tracker (Appendix D).

Q1. Which ion moves through a PEM membrane?
a) OH– **b)** H+ **c)** O2– **d)** K+

Q2. Thermoneutral voltage at 25 °C:
a) 1.23 V **b)** 1.48 V **c)** 2.0 V **d)** 0.9 V

Q3. Charge needed per kg H2 (Faraday):
a) ≈2.66 kAh **b)** ≈26.6 kAh **c)** ≈266 kAh **d)** ≈39.4 kAh

Q4. Which electrolyser uses yttria-stabilised zirconia?
a) AEL **b)** PEM **c)** AEM **d)** SOEC

Q5. PEM anode catalyst per IRENA:
a) Nickel **b)** Iridium oxide **c)** Platinum on carbon **d)** Perovskite

Answer key
Q1 – b: protons; **Q2 – b**: ΔH/(2F); **Q3 – b**: 2F per mol H2; **Q4 – d**: ceramic O2– conductor; **Q5 – b**: IrOx